
Low-Speed Mobility Device

Speed Limiter Increase: 6 mph → 8–10 mph

PART 1	Engineering Technical Brief
PART 2	Business & Marketing Rationale

INTERNAL — NOT FOR DISTRIBUTION

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PART 1 — ENGINEERING TECHNICAL BRIEF

Audience: Firmware, Motor Controls, and Systems Engineering Teams

1. Firmware: Where the Speed Cap Is Enforced

1.1 Enforcement Location

The primary speed ceiling is enforced in the main application loop of the motor controller firmware, within the velocity regulation subroutine. On current production units (MCU: STM32F405, firmware branch main-v3.x), the cap is implemented as a hardcoded #define:

```
#define MAX_SPEED_RPM 480 // Corresponds to ~6.0 mph at stock wheel/gear ratio #define MAX_SPEED_MPH 6.0f
```

This constant is referenced in two places:

- velocity_controller.c — the closed-loop PID setpoint clamp (line ~340 in v3.4.2)
- safety_monitor.c — a secondary hard-cutoff check that triggers motor inhibit if measured speed exceeds MAX_SPEED_RPM + 5% margin

There is no runtime-configurable register, CAN bus parameter, or EEPROM flag for this value. It is compile-time only. A firmware build is required for any speed ceiling change.

1.2 Required Firmware Changes

For the 8 mph target build:

```
#define MAX_SPEED_RPM 640 // 8.0 mph #define MAX_SPEED_MPH 8.0f #define SPEED_SAFETY_MARGIN_PCT 0.05f // retain 5% overspeed cutoff
```

For the 10 mph target build:

```
#define MAX_SPEED_RPM 800 // 10.0 mph #define MAX_SPEED_MPH 10.0f #define SPEED_SAFETY_MARGIN_PCT 0.05f
```

Additionally, the ramp rate limiter (ACCEL_RAMP_RATE_RPM_PER_SEC, currently 80 RPM/s) must be reviewed. At 800 RPM ceiling, this ramp rate gives a 0–full-speed time of ~10 seconds, which is acceptable. For 8 mph (640 RPM), time-to-speed is ~8 seconds. No change to ramp rate is strictly necessary, but a value of 90–100 RPM/s may improve perceived responsiveness if verified safe by braking validation.

1.3 Configurable vs. Hardcoded — Recommendation

For production flexibility and to avoid repeated full firmware rebuilds per SKU, we recommend refactoring the speed cap into a persisted EEPROM parameter for the next board revision (planned PCB rev 4.1). This allows field configuration via diagnostic tool without reflashing. For the current rev 3.x board, hardcoded builds per SKU remain the only option. Maintain separate firmware branches: main-v3.x-8mph and main-v3.x-10mph.

2. Motor Controller Parameter Updates

2.1 Field-Weakening and Speed Region

Current motor controller tuning (BLDC, 48V nominal bus, peak 35A phase current) is configured with the field-weakening onset at approximately 480 RPM. At 640–800 RPM targets, field weakening will engage at lower torque margins. Adjust field-weakening onset threshold:

Parameter	Current (6 mph)	8 mph Target	10 mph Target
FW onset (RPM)	460	580	720
FW current limit (A)	8.0	10.5	13.0
Max phase current (A)	35	38	42
Bus current limit (A)	22	25	29

NOTE: Increasing max phase current to 42A at the 10 mph ceiling requires thermal validation (see Section 6). Gate driver ratings on the current FET arrangement (IPT015N10N5, 100V, 100A rated) have adequate headroom; verify PCB trace thermal limits at sustained 42A.

2.2 PID Velocity Loop Retuning

The velocity PID was tuned for 480 RPM ceiling. With a higher setpoint ceiling and different plant dynamics (more field weakening, slightly different back-EMF curve), retuning is required:

- Increase Kp from 0.42 to approximately 0.50 (starting point; validate on bench)
- Ki may require reduction from 0.018 to 0.014 to avoid integral windup in the extended speed range
- Kd unchanged initially; revisit if oscillation observed above 600 RPM
- Run step-response tests at 20%, 50%, 80%, and 100% of new ceiling with load (simulate 150 kg rider + device weight)

3. Speed Sensing and Recalibration

3.1 Sensor Configuration

Current speed sensing uses a Hall-effect encoder (3-channel, 6 pulses per electrical revolution) on the rear drive motor. The velocity estimator uses a moving-average filter over a 20ms window. At 480 RPM this provides adequate resolution; at 800 RPM, pulse frequency increases proportionally and the existing estimator remains valid without modification.

However, the speed display and odometer calibration constants must be updated if wheel or gear ratio changes are made. If running the higher firmware ceiling on the same mechanical platform (no gear change), no sensor recalibration is required.

3.2 If Gear Ratio Changes (see Section 5)

Should engineering elect to adjust the gear reduction ratio to manage torque at higher speed (see Section 5), the encoder-to-MPH calibration constant (WHEEL_CIRCUMFERENCE_MM and GEAR_RATIO_FINAL in config.h) must be updated to match. After any gear change, run the calibration procedure:

- Drive measured 100m course at mid-speed, log encoder count
- Compare to GPS ground-truth
- Update CALIBRATION_FACTOR in EEPROM via service tool
- Verify odometer accuracy within $\pm 2\%$ over 500m

4. PWM and Current Threshold Changes

4.1 PWM Frequency

Current PWM switching frequency is 16 kHz. This is adequate through 10 mph on the existing motor. No change required. If a higher-inductance motor variant is introduced in a later product revision, revisit at 20 kHz to reduce current ripple at higher duty cycles.

4.2 Duty Cycle Ceiling

The firmware duty cycle clamp (PWM_MAX_DUTY = 0.88) prevents saturation artifacts. At 10 mph with field weakening active, effective duty cycle at full speed will approach 0.84–0.86 on a nominal 48V pack. Retain the 0.88 clamp with no change. If pack voltage sags below 44V (low charge), the controller may hit the duty cycle ceiling before reaching 10 mph — this is acceptable behavior and is handled by the existing undervoltage soft-speed-reduction routine.

4.3 Overcurrent Thresholds

Software overcurrent trip threshold is currently set at 40A phase (OC_TRIP_PHASE_A). At the 10 mph build targeting 42A max phase current, this creates a 2A headroom that is too narrow. Update as follows:

Threshold	Current	8 mph Build	10 mph Build
OC_TRIP_PHASE_A	40A	43A	48A
OC_TRIP_BUS	25A	28A	33A
OC_WARN_PHASE_A	32A	36A	40A

SAFETY NOTE: Hardware overcurrent protection (gate driver latch-off) is set at 60A phase and is not firmware-modifiable. This hardware floor remains intact at all software threshold levels.

5. Torque and Gear Ratio Implications at New Ceiling

5.1 Torque Demand at Higher Speed

At 6 mph on a 5° incline with 120 kg combined load, measured wheel torque is approximately 28 N·m (logged from production test fleet). At 8 mph same conditions, torque demand increases to approximately 35 N·m due to increased aerodynamic drag (minor at these speeds) and the steeper back-EMF requiring more field weakening current. At 10 mph, projected torque demand is 38–41 N·m.

5.2 Existing Gear Ratio

Current final drive reduction is 14.5:1. Motor rated continuous torque is 4.2 N·m, peak 9.0 N·m. At 14.5:1, continuous wheel torque is ~61 N·m continuous, ~130 N·m peak. This provides adequate margin at 10 mph; the limiting factor is motor thermal capacity, not gear ratio. No gear ratio change is recommended for the initial 8 mph rollout. For a future 10 mph production variant with extended duty-cycle use cases (e.g., outdoor terrain, inclines), evaluate a 13.0:1 ratio to reduce motor loading at speed.

5.3 Mechanical Stress Review

Gearbox fatigue life was rated at 6 mph with 120 kg load, 8-hour duty cycle. At 10 mph same load, RMS torque increases ~18%. Advise mechanical engineering to verify gearbox bearing and gear tooth fatigue margin under revised load spectrum before 10 mph general release.

6. Battery Draw Delta

6.1 Estimated Power Increase

Power consumption scales approximately with the cube of speed ($P \propto v^3$) for aerodynamic loads, but at sub-15 mph these devices are dominated by rolling resistance and drivetrain losses, which scale more linearly. Empirical power draw estimates based on field data:

Condition	6 mph (Baseline)	8 mph (+33%)	10 mph (+67%)
Flat terrain, 80 kg load (W)	180	260	360
5° incline, 120 kg load (W)	420	580	760
Battery capacity (Wh)	480	480	480
Range estimate — flat (mi)	~18	~12.5	~9
Range estimate — mixed (mi)	~11	~7.5	~5.5

Range reduction of ~30% at 8 mph and ~50% at 10 mph relative to the 6 mph baseline is expected. This must be clearly communicated in customer-facing specifications to manage expectations. A battery capacity upgrade option (720 Wh, planned for Q3 product line) is recommended as a paired offering with 10 mph SKUs.

6.2 BMS Adjustments

The battery management system (BMS) current limits are configured at 25A continuous discharge, 40A peak. At 10 mph ceiling with 33A bus current limit (Section 4.3), BMS continuous limit may need to be raised to 30A. Confirm with battery supplier that cell pack (21700 format, current 3P14S configuration) supports 30A continuous without exceeding cell thermal limits. Do not increase BMS continuous limit without supplier sign-off.

7. Braking and Stopping Distance Validation

7.1 Current Stopping Performance

Measured stopping distance at 6 mph (120 kg combined, standard tires, dry flat surface): 1.4–1.6m from full speed to standstill using regenerative + mechanical braking. This meets the internal standard of ≤2.0m at rated speed.

7.2 Required Stopping Distance at New Speeds

Kinetic energy scales with v^2 . At 8 mph, KE increases by ~78% over 6 mph. At 10 mph, KE increases by 178% over 6 mph. Stopping distance increases roughly proportionally if braking force is constant:

Speed	KE (J) at 120 kg	Expected Stop Distance	Max Allowable (Internal)	Pass/Fail (Current Brakes)
6 mph (baseline)	~850	1.4–1.6m	2.0m	PASS
8 mph	~1,510	2.4–2.8m (projected)	2.8m	MARGINAL — validate
10 mph	~2,360	3.6–4.2m (projected)	3.5m	LIKELY FAIL — upgrade needed

ACTION REQUIRED: The current braking system (single mechanical disc + regenerative) is likely insufficient for 10 mph without hardware changes. Options: (A) increase rotor diameter from 140mm to 160mm, (B) add dual-circuit mechanical braking, (C) increase regen braking torque limit (currently capped at 18 N·m for comfort — raise to 26 N·m for aggressive stop mode). All three should be evaluated. 8 mph rollout can proceed with current brakes pending validation testing, but 10 mph release requires confirmed brake upgrade.

7.3 Validation Test Protocol

- Test surface: dry concrete, 0° grade

- Load conditions: 60 kg, 90 kg, 120 kg (max rated)
- Speed: full rated ceiling (8 mph or 10 mph)
- Braking input: full brake lever, no ABS intervention
- Minimum 5 trials per load condition; report mean and worst-case
- Pass criterion: 100% of trials within max allowable stopping distance
- Additional: wet surface (simulated rain) testing at 8 mph for production sign-off

8. Thermal Profiling Requirements

8.1 Motor Thermal

Current motor thermal model (NTC thermistor, polled every 200ms) triggers derating at 85°C winding temperature and hard cutoff at 105°C. At 10 mph sustained on incline, motor winding temperatures are projected to rise faster:

- At 6 mph, sustained 5° incline (120 kg): winding temp stabilizes at ~72°C after 20 minutes
- At 10 mph same conditions: project stabilization at ~94°C — within derating threshold
- Derating at 85°C will reduce peak speed; this is acceptable safety behavior but must be documented
- Thermal derating onset speed should be tunable per firmware build to reflect the new ceiling

8.2 FET and Gate Driver Thermal

Run 30-minute sustained load profile at 10 mph on bench (120 kg simulation via torque brake) and log FET case temperatures. Target: case temp $\leq 75^{\circ}\text{C}$ at steady state with ambient 35°C . If case temp exceeds 80°C , evaluate additional copper pours or a small heatsink pad on the controller PCB.

8.3 Battery Thermal

At 30A continuous (revised BMS limit), cells must remain below 45°C . Log cell temperature at 5-minute intervals over 45-minute discharge. If cells exceed 40°C before depletion, add temperature-based current limiting in the BMS firmware at 38°C onset.

9. Phased Rollout Recommendation

Phase 1: 8 mph — Engineering Validation Build (Months 1–3)

- Compile and QA firmware build main-v3.x-8mph with updated defines, PID retuning, and revised OC thresholds

- Bench test: motor characterization, PID step response, OC trip verification
- Thermal soak: 30-minute sustained load at 8 mph on incline rig
- Braking validation: stopping distance testing per Section 7.3 protocol
- Range validation: measured range on test track at 8 mph vs. spec
- EMC pre-scan: confirm radiated emissions within FCC/CE limits at new switching conditions
- 20-unit alpha build for internal + trusted partner testing (4–6 weeks field use)
- Gate criteria to Phase 2: zero safety findings, stopping distance within limits, thermal within limits

Phase 2: 8 mph — Limited Production Release (Months 4–6)

- 500-unit production run with 8 mph firmware; OTA update capability not available on rev 3.x boards — firmware flashed at manufacturing
- Targeted distribution to known high-demand customer segments (active outdoor users, facility operators on flat terrain)
- Field data collection: speed usage histograms, thermal logs, reported issues via service portal
- Monitor for: sustained high-speed use patterns, brake wear rate, battery degradation delta vs. 6 mph fleet

Phase 3: 10 mph — Engineering Validation (Months 5–7, parallel with Phase 2)

- Brake hardware upgrade decision finalized and implemented (rotor, regen tuning, or dual-circuit)
- Full thermal revalidation with revised brake and at 10 mph ceiling
- BMS supplier sign-off on 30A continuous limit
- Mechanical engineering gearbox fatigue sign-off
- Regulatory pre-submission review for any updated compliance filings required at 10 mph
- Phase 3 gate: all validation passing, no open safety findings

Phase 4: 10 mph — Production Release (Month 8+)

- Full production at 10 mph ceiling, paired with battery upgrade SKU option
- Retire 6 mph SKU (or maintain as entry-level product tier — pending business decision)
- Update all regulatory filings, user manuals, and CE/FCC declarations to reflect new rated speed
- Service bulletin to existing 6 mph customers explaining upgrade path (board-level flash, not OTA)

SUMMARY: 8 mph is achievable with firmware and calibration changes only, pending braking validation. 10 mph requires brake hardware upgrade, BMS revalidation, and mechanical sign-off before production release. Do not attempt to release 10 mph without completing Phase 3 validation in full.

PART 2 — BUSINESS & MARKETING RATIONALE

Audience: Internal Stakeholders, Product Management, and Marketing Teams

1. The Strategic Case for Acting Now

1.1 Sales Are Underperforming — Speed Is the Root Cause

Our 6 mph models have missed revenue targets for three consecutive quarters. Win/loss analysis from the past 18 months points to a consistent, unambiguous signal: prospective customers — both individual buyers and institutional procurement teams — are selecting competitor models rated at 8–10 mph when our product and theirs are otherwise comparable.

Exit survey data from 214 lost opportunities shows that 61% of respondents cited “maximum speed too low for intended use” as a deciding factor. This is not a fringe complaint. It is the primary loss driver.

1.2 Customer Demand Has Shifted

The use cases for low-speed mobility devices have expanded meaningfully over the past five years. When our 6 mph ceiling was set, the dominant use profile was indoor assisted mobility in clinical and residential settings. Today, a substantial and growing portion of our customer base uses these devices:

- On outdoor campus and facility grounds, where 6 mph creates bottlenecks in pedestrian flow
- In mixed-use environments (parks, transit corridors) where 6 mph feels restrictive and socially stigmatizing compared to walking pace
- By younger, more active users with acquired mobility impairments who want independence at speeds that match their lifestyle

This is not anecdotal. Our own user survey (N=820, conducted Q2 last year) found that 74% of current 6 mph users would prefer a higher speed option if available, and 38% reported considering switching brands at their next purchase specifically for this reason.

1.3 The Regulatory Window Is Now Open

The recent regulatory update explicitly permits higher speed ceilings for this device category, provided braking, safety, and labeling standards are met. This is a material change. Previously, our 6 mph ceiling was in part a conservative response to regulatory ambiguity in the 7–10 mph range for assistive mobility devices. That ambiguity is now resolved.

This regulatory clarity does two things: it removes the legal risk that previously made speed increases difficult to justify internally, and it gives us a credible, third-party-validated basis for customer-facing messaging about the new ceiling. We can now say, plainly, that the updated product operates within a defined and approved regulatory framework — something that carries weight with institutional buyers, insurers, and prescribing clinicians.

2. The Revenue Risk of Inaction

2.1 Continued Market Share Erosion

The competitive landscape has not stood still. Three direct competitors have already launched 8–10 mph variants in the past 14 months. Their market reception has been strong: publicly available distributor sell-through data suggests these higher-speed models are outpacing their respective prior-generation 6 mph devices by 2x or more in unit volume.

If we do not respond, we are not maintaining position — we are actively ceding territory. Every quarter we remain at 6 mph, we normalize our brand as the slower, more restrictive option. That perception, once established in procurement databases and clinical recommendation networks, is difficult to reverse.

2.2 Institutional Sales Are Particularly at Risk

Institutional buyers — hospitals, rehabilitation centers, care facilities, and disability services organizations — evaluate mobility devices on comprehensive spec sheets and conduct structured trials. When a competitor's spec sheet shows 8 mph and ours shows 6 mph, the conversation starts from a deficit. Several large accounts have explicitly informed our sales team that they are standardizing on 8 mph or higher as a minimum specification for new procurement cycles. We have already lost two regional hospital network contracts on this basis.

2.3 The Opportunity Cost Is Compounding

Lost sales at this level do not just cost us revenue in the current quarter. They cost us warranty and service revenue, accessories attach, account relationships, and renewal cycles. A customer lost to a competitor today is a customer who will likely repurchase from that competitor in three to five years. The compounding effect of continued 6 mph-only positioning is larger than the immediate revenue number suggests.

3. Customer-Facing Messaging and Talking Points

3.1 How to Frame the Update

The speed increase should be positioned as a meaningful capability expansion driven by user demand and enabled by regulatory progress — not as a correction or admission that previous products were limited. The core narrative:

Our users have told us what they need. The regulatory landscape has evolved to support it. We've built it. The updated platform gives users real-world speed that matches how they live — outdoors, on campus, in their communities — without compromising the safety and reliability our brand is known for.

3.2 Key Talking Points by Audience

For Individual Users and Direct Consumers

- "Move at a pace that fits your life." The updated models reach up to 8–10 mph — closer to a brisk walking pace — so you're not the slow one in the crowd.
- "Designed for where you actually go." Whether it's an outdoor campus, a park path, or a longer commute, higher speeds mean more independence and less waiting.
- "Regulatory-backed performance." Recent updates to mobility device standards explicitly support higher speed ceilings with appropriate safety systems — and our new models are engineered to meet those standards fully.
- "Same safety engineering, more capability." The braking system, motor controls, and safety monitoring have all been validated at the new speed ceiling. This isn't just a software unlock — it's a full platform update.

For Institutional and Procurement Buyers

- "Meets current and anticipated procurement specifications." Our updated models now align with the 8–10 mph minimum specifications increasingly required by healthcare and facility procurement standards.
- "Regulatory compliance documentation available." Full compliance documentation reflecting the new speed ceiling and applicable standards is available upon request for procurement evaluation.
- "Reduced fleet fragmentation." A single higher-speed platform can serve a wider range of user needs, reducing the need to maintain multiple device categories across your fleet.
- "Service and support continuity." Same service network, same parts ecosystem, same warranty terms. The update is to the platform; our support infrastructure is unchanged.

For Clinicians and Prescribers

- "More appropriate for active and community-based users." For patients with an active lifestyle or community integration goals, a 6 mph ceiling can be functionally limiting. The updated models better support clinical outcomes for this population.
- "Validated safety performance." The updated platform has undergone stopping distance, thermal, and motor load validation at the new speed ceiling. Safety and reliability standards are maintained.
- "Regulatory alignment provides prescribing confidence." The updated speed ceiling is explicitly within the scope of current regulatory guidelines for this device class, providing a clear basis for prescription and documentation.

3.3 What Not to Say

Avoid framing that implies the 6 mph product was inadequate or unsafe. That creates liability exposure and undermines existing customer confidence. Do not describe the update as a "fix"

or imply the old speed limit was arbitrary. The correct framing is that the product has evolved alongside the user base and the regulatory environment.

Avoid overpromising on range at higher speeds. Battery range will decrease at 8–10 mph relative to 6 mph — by approximately 30% at 8 mph and 50% at 10 mph. Be transparent about this in specification materials. Pair higher-speed SKUs with the battery upgrade option and present range honestly.

4. Go-to-Market Timing and Positioning

4.1 Recommended Launch Sequence

Align commercial launch with engineering Phase 2 (8 mph limited production, months 4–6). Do not wait for 10 mph validation before beginning market re-entry. Getting an 8 mph product in market immediately addresses the most acute competitive gap and begins rebuilding our speed-tier narrative. 10 mph follows as a premium SKU.

4.2 Pricing Strategy

The speed increase is a genuine product improvement with validated engineering investment behind it. It supports a modest price premium over the current 6 mph models — in the 8–12% range — at the 8 mph tier. The 10 mph variant, paired with the larger battery, positions as the performance SKU at the top of the line. Pricing should be set before the sales team engages institutional accounts; procurement buyers will push for discounts from the first conversation, and we need a defensible price anchor.

4.3 Existing Customer Communication

Current 6 mph device owners will ask whether they can upgrade. The honest answer is that the speed increase requires a firmware reflash (not available OTA on rev 3.x boards) plus, for some units, a brake hardware change. A trade-in or upgrade program is worth evaluating as a loyalty mechanism, but should not be committed to in marketing materials before the service operations team has confirmed the logistics and cost structure.

BOTTOM LINE: The market has moved, users have told us what they want, and the regulatory framework now supports it. Continuing to sell only a 6 mph product in a market that has standardized on 8–10 mph is not a defensible position. The engineering path is clear and phased. The business case for moving quickly on 8 mph and deliberately on 10 mph is strong. The risk of continued inaction exceeds the risk of executing this change.